

# Deep Learning-Based Biosignal Control of a Soft Hand Exoskeleton for Grasp Assistance

## Supplementary Information

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*This Supplementary information includes:*

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## S1. EEG Channel Selection and Spectral Analysis

This section details the EEG spectral analysis and justifies the selection of the central and frontal regions across the delta–beta frequency bands to identify which channels and frequency bands contained the most discriminative information. The Power Spectral Density (PSD) was computed using the processed EEG method according to Section 2.6, EEG preprocessing. However, for this specific analysis, the bandpass filter was extended to 0.5 – 60 Hz to capture the full spectrum of interest. Following the international 10 – 20 system, the electrodes were grouped into five regions: pre-frontal (Fp1, Fp2), frontal (F7, F3, Fz, F4, F8), central (C3, Cz, C4), temporal (T3, T5, T6, T4), partial (P3, P4). The analyzed frequency bands included delta (0.5 – 4 Hz), theta (4 – 8 Hz), alpha (8 – 13 Hz), beta (13 – 30 Hz), gamma (30 – 60 Hz). For each participant, the PSD was computed across all trials and grouped by class ( $c \in [\text{rest, onset, offset}]$ ), region ( $r$ ), and frequency band ( $f$ ). The absolute band power ( $E_{c,r,f}$ ), was obtained by integrating the single trial PSD using the trapezoidal rule. The absolute band power was normalized to reduce inter-subject variability, which we will denote as relative power ( $E_{\text{relative}}$ ). The relative power was calculated by dividing the specific band power by the total power across the entire spectrum of frequencies (0.5 – 60 Hz):

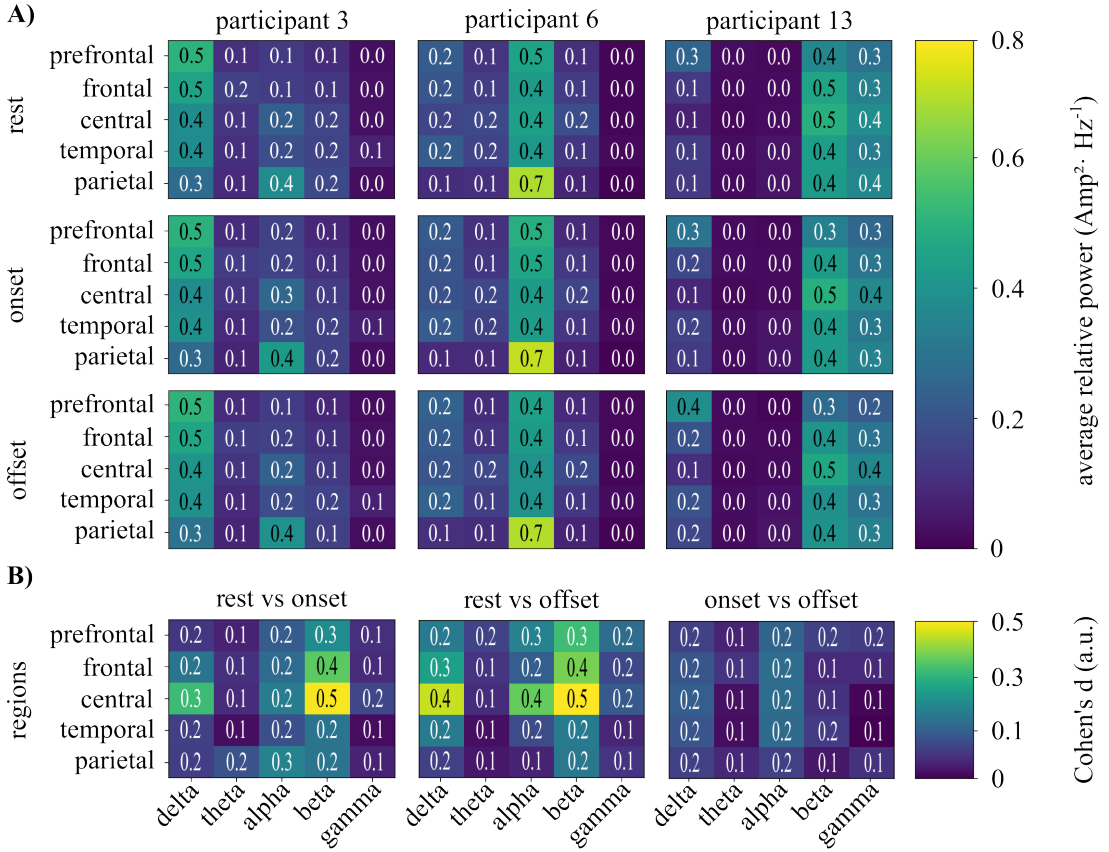
$$E_{\text{relative}} = \frac{E_{c,r,f}}{E_{c,r,0.5-60}} \quad (\text{S1})$$

Finally, the relative power (shown in Eq S1) were averaged across all trials within each class to obtain a single robust metric per participant, region and frequency. The average relative power is measured as amplitude squared per unit of frequency and can be observed in the Figure S1A mesh grid. These participants was chosen to highlight inter-subject variability. However, a general observation regarding all 17 participants, showed that the highest relative power appeared in prefrontal region for delta band, alpha in parietal region, and beta in central region.

To investigate which regions and frequency bands contribute with strong class separation, Cohen’s  $d$  was calculated, as defined in Eq S2:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 \cdot (n_1 - 1) + s_2^2 \cdot (n_2 - 1)}{n_1 + n_2 - 2}}} \quad (\text{S2})$$

where,  $\bar{x}_1$  and  $s_1$  are the estimated mean and standard deviation of the relative power across trials at one specific region and frequency band in the first class, whereas  $\bar{x}_2$  and  $s_2$  is the equivalent metrics for a other class. The variables  $n_1$  and  $n_2$  denote the corresponding number of trials per class. The Cohen’s  $d$  formula were computed across all participants and averaged to yield a global class separation influenced by all participants. Figure S1B illustrates these Cohen’s  $d$  values. The result indicates the strongest class separation belong within the central and frontal region across delta, alpha, and beta frequency bands for both the rest vs onset and rest vs offset conditions. However, a weak class separation appears in onset vs offset condition. Indicating how the relative power remain similar between the initialization of a finger flexion motion and its return to an idle state.

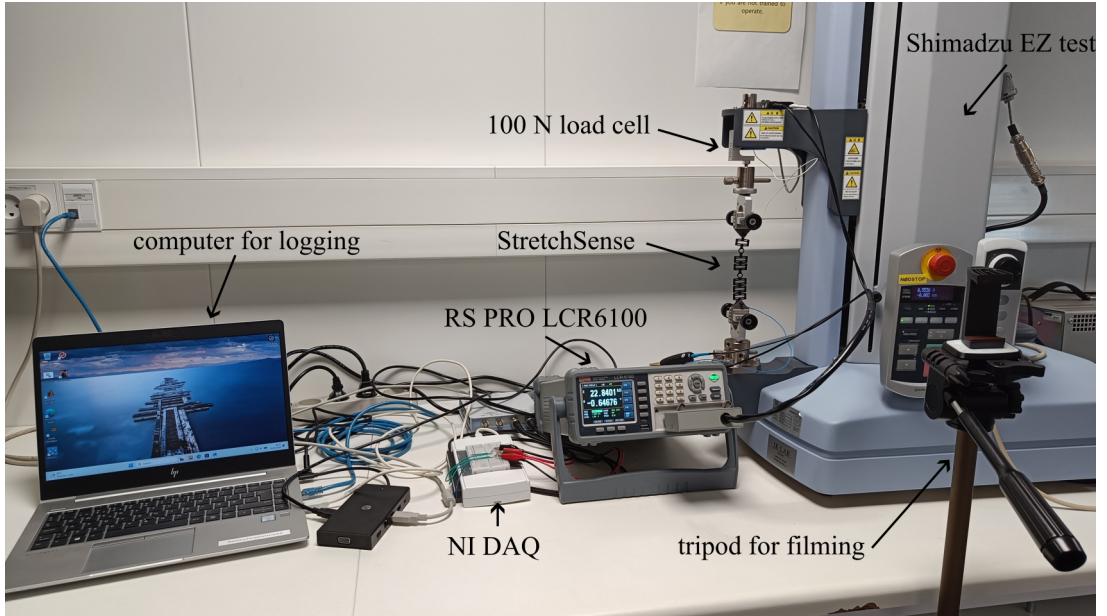


**Figure S1.** **A** shows the power density spectrum (PSD) across regions (prefrontal, frontal, central, temporal, and parietal) and frequency bands (delta, theta, alpha, beta, and gamma) for three participants with strong dissimilarity in PSD. Each row present the classes of index and thumb onsets, offsets, and resting state. **B** displays class separation (rest vs onset, rest vs offset, and onset vs offset) using Cohen's method, based on the PSD for all 17 participants across regions and frequency bands..

While individual participants demonstrates variations in strongest relative brain power at different regions and frequency bands, the Cohen's d analysis demonstrates that the highest class separation occurs when comparing the rest state against active states. This distinct separation is primarily located within the central and frontal regions of the brain across the delta, alpha, and beta frequency bands (0.5 – 30 Hz). With that in mind, restricting the channels to only include central and frontal region within the 0.5 – 30 Hz range represents a feasible reason for channel and frequency band selection. Importantly, two disadvantages is noted using this methodology. Firstly, the Cohen's d evaluates the magnitude of statistical difference between classes, excluding those regions and frequency with highest relative power.

Secondly, while PSD effectively computes overall energy across frequency components, it does not capture time-locked responses nor specific oscillatory patterns due to stimuli. Which could display an alternative picture of EEG behavior.

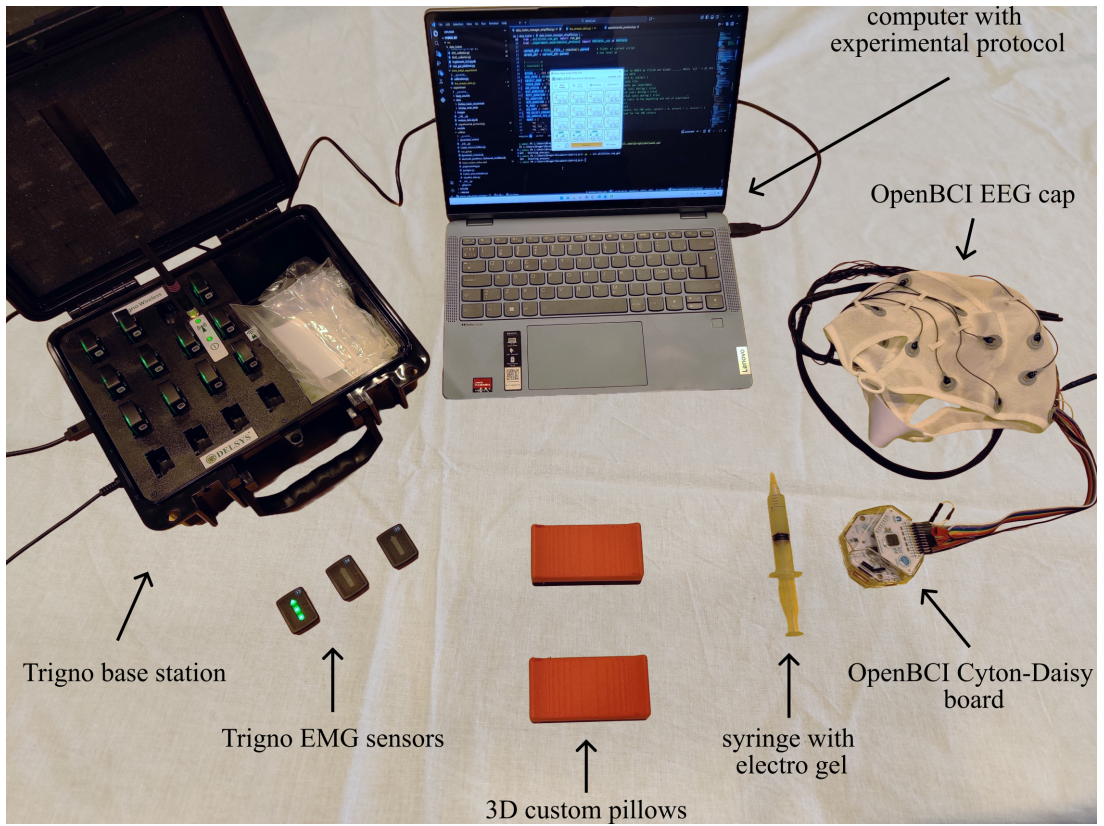
## S2. Experimental Setup for Characterization of StretchSense



**Figure S2.** This shows the experimental setup for cyclic testing of the StretchSense sensor. For repeatability and high precision, the Shimadzu EZ test with a 100 N load cell was used as the tensile tester. StretchSense was mounted using two 3D printed clamps designed for the EZ test, as was held securely in place, with little to no lateral movement. From StretchSense a wire is attached at each end, at these the RS PR LCR6100 is attached with alligator clips to measure the resistance change during the testing. The clips are mounted on the testing machine in a way to not transfer any load to the load cell or the sensor, skewing the results. The LCR meter is set to output impedance  $Z$  in  $\Omega$  and the angle in degrees, and logging in fast mode. The LCR meter is connected to the computer via a USB cable. The EZ test machine outputs the force and the stroke distance in an analog voltage between  $-5\text{ V}$  and  $5\text{ V}$ , where the range these voltages represent has been set individually for each test, to get the highest resolution. The analog signals are captured by the National Instruments DAQ (NI DAQ) that samples and converts the analog voltage to digital signal, that is transmitted to the computer via a USB cable. The computer is running a custom MATLAB script, which is responsible for synchronizing all the incoming data and log it for use afterwards. It also runs the software called Trapezium X, needed to set up and run the cyclic tests on the testing machine.



### S3. Experimental Setup for Index and Thumb Classification



**Figure S3.** This outlines the hardware components for index and thumb classification experiment. EMG acquisition required three Trigno Avanti<sup>TM</sup> sensors, which transfers muscular activation wirelessly to Trigno base station. The OpenBCI EEG cap allows for non-invasive electrical activity recordings at 19 locations on the scalp, following the 10-20 system. The Cyton-Daisy board allows for amplifying the EEG for 16 channels and transfer these via a USB dongle. The syringe with electro gel is used to mitigating scalp impedance levels and apply before experimental acquisition. During experimental protocol, the user places their wrist and elbow on two 3D printed varioShore pillows. The computer executes the experimental protocol and continuously collects data via the USB A from the Trigno base station and the OpenBCI Cyton-Daisy board via a USB dongle..

#### **S4. Supplementary Videos**

Supplementary Video S1: Characterization of Integrated Exoskeleton Sensors.

Rich media available at <https://youtu.be/mhyjGOVmtmI>

Supplementary Video S2: Object Grasping Using Index Module.

Rich media available at <https://youtu.be/S3xFVID3CKw>

Supplementary Video S3: Index and Thumb Flexion Classification Experimental Protocol.

Rich media available at <https://youtu.be/pHsXzFFM41g>

Supplementary Video S4: Demonstration of Real-Time Hand Motion Classification.

Rich media available at <https://youtu.be/5kJ1KMPTAkE>

Supplementary Video S5: EMG-Based Evaluation of Hand Exoskeleton Assistance and Muscle Effort.

Rich media available at <https://youtu.be/5mZSDJYYHPM>

Supplementary Video S6: Demonstration of Real-Time Model Control of Pinch Capable Exoskeleton.

Rich media available at <https://youtu.be/9jlsU56XesI>

## **S5. GitHub**

The following link points at the GitHub repository that contains the files used for the development of the soft hand exoskeleton and the deep learning-based biosignal classification framework for real-time grasp assistance. All design files, code, and more practical information is found there.

Repository available at <https://github.com/MegaMagn/dl-biosignal-sofie-control.git>

S6. Contributions

*Table S1. Contributions.*

| Task                                         | Nicklas | Magnus |
|----------------------------------------------|---------|--------|
| Experiments                                  | 50%     | 50%    |
| Writing                                      | 50%     | 50%    |
| Hardware design of exoskeleton               | 0%      | 100%   |
| Characterization of integrated sensors       | 0%      | 100%   |
| Deep learning based control framework        | 100%    | 0%     |
| Training and validation of models            | 100%    | 0%     |
| Conference article SAB 2026 (in peer-review) | 60%     | 40%    |
| Conference article CBS 2026 (in peer-review) | 10%     | 90%    |